



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 1 REPORT

Internship Program — Batch 2026

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1. Introduction

This report presents the work completed for **Task 03** of the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. The internship focuses on practical exposure to core concepts in Artificial Intelligence and Machine Learning, equipping interns with skills applicable to real-world technology and marketing challenges.

M-Tech Production & Marketing is a forward-looking company based in Haripur, KPK, specializing in Technology, Training, Marketing, and Projects. The internship is structured in a 9-Phase roadmap designed to progressively build theoretical knowledge and hands-on project experience.

2. Objectives

- Strengthening the fundamentals of Python programming required for AI and Machine Learning development.
- Develop a clear understanding of variables, data types, operators, and type conversion.
- Learn string manipulation techniques and basic user input handling.
- Practice decision-making using conditional statements (if, elif, and else).
- Implement iteration using for and while loops to solve programming problems.
- Understand and manipulate Python data structures such as lists, tuples, dictionaries, and sets.
- Create reusable functions to improve code organization and reduce repetition.

3. Task Description

The task involved implementing the foundational modules of the AI Lab Manual. This included:

- Developing programs for user input and type conversion.
- Implementing conditional logic and iteration using for and while loops.
- Manipulating data structures like Lists and Dictionaries.
- Defining modular functions to perform repetitive calculations.

4. Work Done

- **Python Foundations:** Implemented programs for user input, type conversion, arithmetic operations, and string manipulation.
- **Conditional Statements:** Developed programs using if, elif, and else statements to solve different logical problems.
- **Looping Concepts:** Implemented for and while loops to generate multiplication tables, identify prime numbers, and perform repetitive calculations.
- **Data Structures:** Practiced creating and manipulating lists, tuples, dictionaries, and sets to organize and process data efficiently.

- **Functions:** Created reusable Python functions for tasks such as factorial calculation, grade evaluation, and other repetitive operations.
- **Jupyter Notebook Practice:** Completed all exercises within Jupyter Notebook while documenting outputs and understanding the execution flow of each program.

5. Challenges Faced

- Managing variable scope and logical conditions while implementing the prime number program.
- Understanding the correct use of nested loops and boolean flags for accurate program execution.
- Selecting the most suitable data structure for different programming tasks.
- Debugging syntax and logical errors during function creation and loop implementation.

6. Key Learnings

- Developed a strong understanding of Python syntax and programming fundamentals.
- Learned how modular programming using functions improves code readability, reusability, and maintenance.
- Gained practical experience working with Python data structures and understood when each should be used.
- Improved problem-solving skills by implementing conditional statements and iterative loops.
- Learned the effective use of break and continue statements to control loop execution.
- Built a solid programming foundation that will support future AI, Machine Learning, and Data Science projects.

7. Conclusion

This task successfully strengthened my understanding of Python programming fundamentals and improved my confidence in solving programming problems using logical thinking. By completing exercises on variables, control flow, data structures, and functions, I developed essential skills required for Artificial Intelligence and Machine Learning applications. The knowledge gained from this task has prepared me to work on more advanced algorithms and practical AI projects in the upcoming phases of the internship.

Intern Signature: Hamza Badshah

Name: Hamza Badshah

Date: 31 June, 2026

Supervisor: _____

Name:

Designation: